

Practice Questions: Operational Analysis Solution

1. Consider the mail server of a company where the employees make intense use of e-mail services. The mail server processes about 35,000 requests (e.g., sendmail) on a working day. However, the mail service requests are not uniformly spread over the 24-hour and there is a concentration of requests during a peak hour. The average inter-arrival time of mail service requests during peak hour is 0.80 seconds.

On average, the SMTP (i.e., Simple Mail Transfer Protocol) service processes 2.5 messages per second.

- a. What is the **average response time** during the peak hour?

$$\lambda = 1/0.8 = 1.25 \text{ messages/s}$$

$$\mu = 2.5 \text{ messages/s}$$

$$\text{utilization } (u) = \lambda/\mu = 0.5$$

$$\text{response time} = s/(1-u) = 0.8\text{s}$$

- b. From measurements taken of the mail server, we observe that doubling the mail size increases the service time by 40%. What is the average response time if the average mail size is twice as big?

Double mail size and increase service time by 40%,

$$s = 1/\mu \times 1.4 = 0.56\text{s}$$

$$u = 1.25 \times 0.56 = 0.7$$

$$R = 0.56/(1-0.7) = 1.87\text{s} \quad \uparrow 134\%$$

- c. What is the average response time if the mail server is replaced with a server that is twice as fast?

$$s = (1/2.5)/2 \quad \text{server twice as fast}$$

$$s' = 0.2$$

$$u = 1.25 \times 0.2 = 0.25$$

$$R = 0.2/(1-0.25) = 0.27\text{s} \quad \downarrow 66\%$$

- d. What is the average response time if the average mail size is doubled when the mail server becomes twice as fast? Compare the average response time with that obtained in (b) and (c).

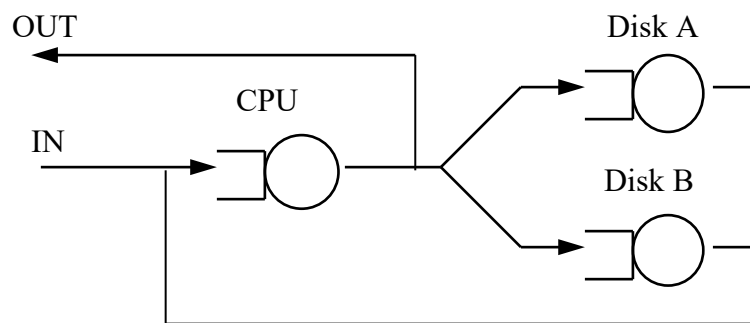
combining (b) and (c)

$$s = 0.56/2 = 0.28\text{s}$$

$$R = 0.28/(1-0.35) = 0.43\text{s} \quad \downarrow 46\%$$

2. A queuing model of a file server consists of a CPU and two disks, A and B. Measurements on a distributed system with *six-client* system making requests to the file server produced the following data:

Observation interval = 3600 sec
 No. of client requests = 10,800
 CPU busy time = 1728 sec
 Disk A busy time = 1512 sec
 Disk B busy time = 2592 sec
 No. of visits (I/O requests) to disk A = 75,600
 No. of visits (I/O requests) to disk B = 86,400



Compute the device utilizations, device response times and server response time.

X (throughput) = $10800/3600 = 3$ client requests / second

$V_A = 75600/10800 = 7$ visits per client request to disk A

$V_B = 86400/10800 = 8$ visits per client request to disk B

$V_{CPU} = 1+7+8 = 16$ {note how this is derived}

$D_{CPU} = 1728/10800 = 0.16$ seconds of CPU time per client request

$D_A = 1512/10800 = 0.14$

$D_B = 2592/10800 = 0.24$

$S_{CPU} = 0.16/16 = 0.01$ second per visit to CPU

$S_A = 0.14/7 = 0.02$

$S_B = 0.24/8 = 0.03$

$U_{CPU} = X D_{CPU} = 3 \times 0.16 = 0.48$

$U_A = 3 \times 0.14 = 0.42$

$U_B = 3 \times 0.24 = 0.72$

$R_{CPU} = S_{CPU}/(1 - U_{CPU}) = 0.0192$ seconds

$R_A = 0.02/(1 - 0.42) = 0.0345$ seconds

$R_B = 0.03/(1 - 0.72) = 0.107$ seconds

$\therefore R = 1.406$ seconds

Quantify the impact of the following changes:

- a. increase the no. of clients to 8;

We assume that the new clients make requests similar to those measured. The request arrival rate will go up by a factor of 8/6, thus,

$$X = 4 \text{ requests / second}$$

Analysis:

$$\begin{aligned} U_{\text{CPU}} &= X D_{\text{CPU}} = 4 \times 0.16 = 0.64 \\ U_A &= X D_A = 4 \times 0.14 = 0.56 \\ U_B &= X D_B = 4 \times 0.24 = 0.96 \\ R_{\text{CPU}} &= S_{\text{CPU}} / (1 - U_{\text{CPU}}) = 0.01 / (1 - 0.64) = 0.0278 \text{ seconds} \\ R_A &= 0.0455 \\ R_B &= 0.75 \\ R &= 16 \times 0.0278 + 7 \times 0.0455 + 8 \times 0.75 = 6.75 \text{ seconds} \end{aligned}$$

Thus, if the number of clients is increased from 6 to 8, the server response time will degrade by a factor of $6.75/1.406 = 4.8$

- b. use a cache for disk B with a hit ratio of 50%, although it increases the CPU overhead by 30% and the disk B service time (per I/O) by 10%;

use a cache for disk B with a hit rate of 50%.

$$\therefore V_B = 0.5 \times 8 = 4$$

$$\begin{aligned} &\text{Increase CPU overhead by 30\%,} \\ S_{\text{CPU}} &= 1.3 \times 0.01 = 0.013 \end{aligned}$$

$$S_A = 0.02$$

$$\begin{aligned} &\text{Increase disk B service time by 10 \%,} \\ S_B &= 1.1 \times 0.03 = 0.033 \end{aligned}$$

Analysis:

$$U_{\text{CPU}} = X D_{\text{CPU}} = 3 \times 12 \times 0.013 = 0.468$$

Since there is a cache for Disk B, fewer jobs go from CPU to Disk B. The 4 visits that experience a cache hit would continue their processing on the CPU instead of going to Disk B. Therefore, there is increased service time on the CPU (by 30% because the request spends more time on the CPU) and increased service time at disk B. This means that overall, V_B decreases to 4, making the V_{CPU} decrease to 12 (instead of the initial 16).

Specifically, the jobs must visit the CPU before going to disk A or disk B and one last time before exiting the system. As such, a request has a visit ratio to CPU of 7 (every time before going to disk A) + 4 (every time before going to disk B) + 1 (just before exiting the system through OUT).

Since the visit ratio at disk B changes, the visit ratio at the CPU will change (you may picture it as the request travels less throughout the system and gets longer processing times at the CPU and disk B when the cache is added).

$$\begin{aligned}
 U_A &= 3 \times 7 \times 0.02 = 0.42 \\
 U_B &= 3 \times 4 \times 0.033 = 0.396 \\
 R_{CPU} &= S_{CPU} / (1 - U_{CPU}) = (0.013) / (1 - 0.468) = 0.013 / 0.532 = 0.0244 \text{ seconds} \\
 R_A &= 0.02 / (1 - 0.42) = 0.02 / 0.58 = 0.0345 \\
 R_B &= 0.033 / (1 - 0.396) = 0.033 / 0.604 = 0.0546 \\
 R &= 0.0244 \times 12 + 0.0345 \times 7 + 0.0546 \times 4 \\
 &= 0.2928 + 0.2415 + 0.2184 \\
 &= 0.7527 \text{ seconds}
 \end{aligned}$$

Thus, if we use a cache for disk B, then server response time will improve by $(1.406 - 0.7527) / 1.406 = 46\%$

- c. have a lower cost server with only one disk (disk A) and direct all I/O requests to it.

We assume that there is enough space on disk A to be able to accommodate all files.

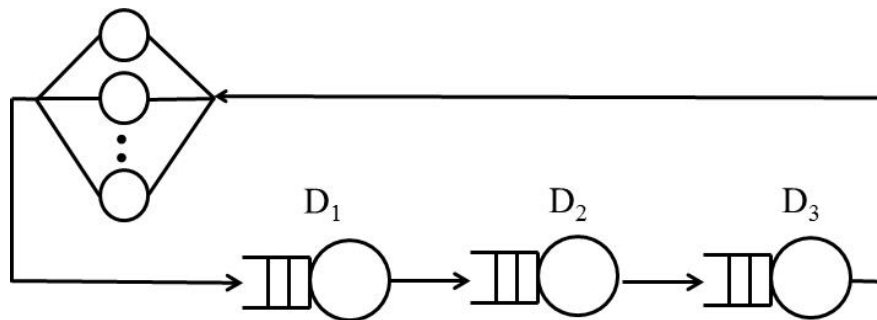
Analysis:

$$\begin{aligned}
 V_B &= \emptyset \\
 V_A &= 7 + 8 = 15 \\
 D_{CPU} &= 0.16 \\
 D_A &= 15 \times 0.02 = 0.3 \\
 U_{CPU} &= 3 \times 0.16 = 0.48 \\
 U_A &= 3 \times 0.3 = 0.9 \\
 R_{CPU} &= 0.0192 \\
 R_A &= 0.2 \\
 R &= 16 \times 0.0192 + 15 \times 0.2 = 3.31 \text{ seconds}
 \end{aligned}$$

Thus, if we replace the two disks by one, the previous response time will degrade by a factor of $3.31 / 1.406 = 2.35$

[note how we derived the visit ratios]

3. The figure below shows a computer system with three subsystems. Based on measurement, the service demands of the three tandem service centers (subsystems) are $D_1 = 2$ seconds, $D_2 = 3$ seconds and $D_3 = 1$ second.



- a. What is the bottleneck device? What is the maximum achievable system throughput?

$D_1 = 2$ sec

$D_2 = 3$ sec

$D_3 = 1$ sec

D_2 with the highest service demand is the bottleneck

Max throughput is bound by bottleneck device

$$X_{\text{sat}} = 1/D_{\text{max}} \\ = 1/3 \text{ jobs per sec}$$

- b. What is the best system response time?

Best response occurs when there is no queueing

$$R_{\text{min}} = D_1 + D_2 + D_3 = 6 \text{ sec}$$

- c. What is the worst system response time? Consider two cases, if the system is an open system and a closed system.

Worst response time: For open system, there is no worst time. Beyond the saturation point, R continues to get worse as the load increases. For closed system, the limit is determined by the job population N .

- d. What are the asymptotic and balanced system bounds for system response time and system throughput? Comment on your results.

Asymptotic bounds:

$$D = D_1 + D_2 + D_3 = 6$$

$$D_{\max} = 3$$

$$R(N) \geq \max(6, N * 3)$$

$$X(N) \leq \min(1/3, N/6)$$

$$N^* = 6/3 = 2 \text{ terminals}$$

Balanced bounds: load spread on all devices.

$$D_{\text{avg}} = 6/3 = 2 \quad D_{\max} = 3$$

$$\frac{N}{6+(N-1)3} \leq X(N) \leq \min\left(\frac{1}{3}, \frac{N}{6+(N-1)2}\right)$$

$$\frac{N}{6+(N-1)3} \leq X(N) \leq \min\left(\frac{1}{3}, \frac{N}{(N+2)2}\right)$$

$$\max(N6, 6 + (N - 1)2) \leq R(N) \leq 6 + (N - 1)3$$